

Bridge Test and Redesign Evidence

NAME _____ DATE _____ PERIOD _____

Fixed Dataset — Complete Default Route

All values below are fictional classroom evidence using the same staged load and stop rule.

SAMPLE	VISIBLE SYSTEM	MAXIMUM COMPLETED STAGE	STOP EVIDENCE
A	Flat beam; two main rails; taped card deck	2	Center deck reached displacement mark before Stage 3
B	Triangulated side frames; cross-braced deck	5 safe cap	No detached joint or displacement stop at cap
C	Curved/segmented sides; narrow support contact	3	One support contact slipped before Stage 4

Before revealing the results, predict which sample will complete the highest stage and cite one visible clue:

Optional demonstration boundary: A teacher may demonstrate one prepared model only after a tested kit, secured supports, catch tray, marked keep-clear zone, known lightweight stages, safe cap, objective stop rule, and workable class schedule are ready. Students do not place loads. Never use textbooks or unstable desks.

Reveal and Compare

Sample I analyzed: [A / B / C]

Maximum completed stage: [] Stop evidence: []

First failed or limited element: []

Which comparison is fair, and what variable is still uncontrolled?

One pattern and one exception across the three results:

Word bank: stage · stop evidence · displacement · support contact · variable · limitation

Individual Failure Analysis and Redesign

Result evidence I used:

One specific redesign:

Why that change should affect the result:

What I would measure in the next test:

Use this frame: Sample [code] reached [stage/result]. I would change [specific feature] because [evidence]. In the next test, I would measure [measure].

Career Role and Real-World Limits

Worker who would review this result next:

Additional evidence a real project would require:

One limit of the classroom prototype:

Use this frame: A [role] would review [evidence] next, but a real project would also require [additional evidence].
This classroom model cannot prove [limit].